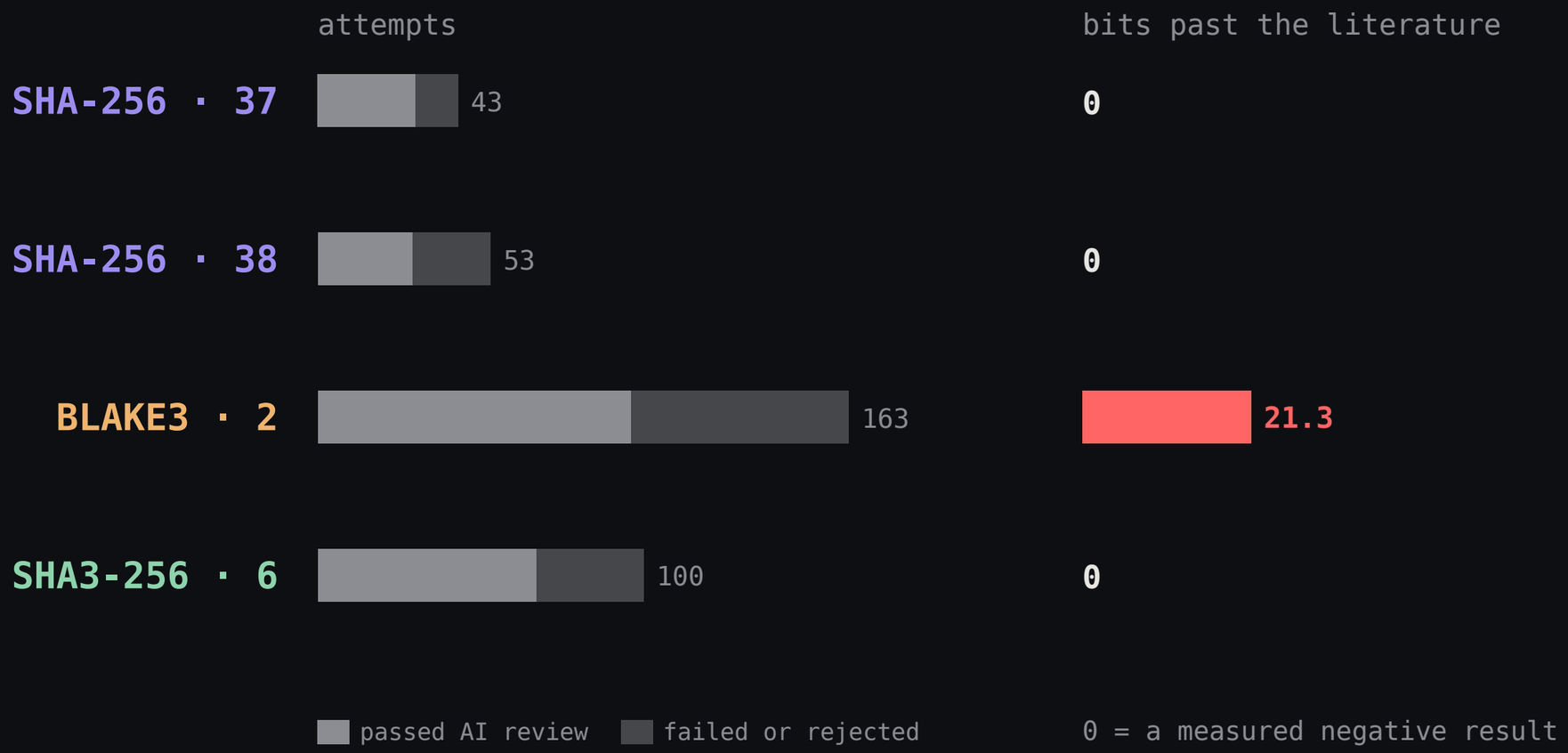
backup · **Negative results, measured**



backup · Secure hashes: necessary, sufficient, strictly safer

necessary

RSA · ECDSA · Ed25519 · ML-DSA · TLS · git · Bitcoin

secure hash

✗ public-key encryption

sufficient

- **signatures** — SLH-DSA
- **MACs** — HMAC
- **KDFs** — HKDF
- **random bits** — DRBG
- **encryption**
- **commitments** — Merkle
- **proofs** — STARKs

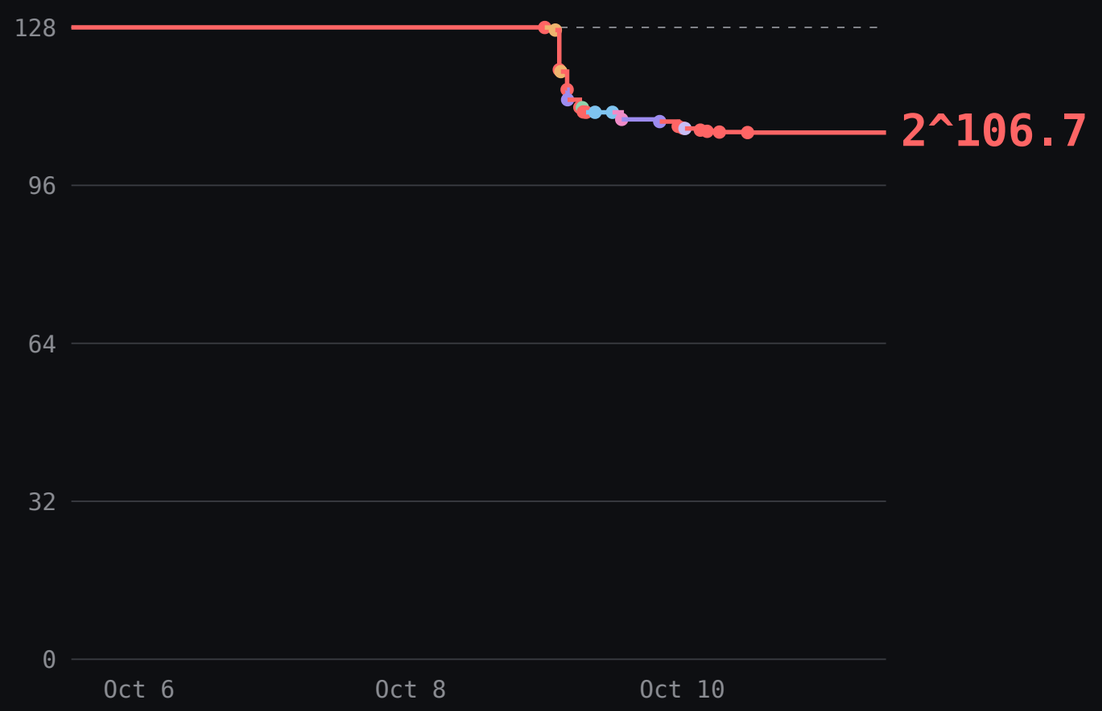
strictly safer



backup · One week of HashSmash

BLAKE3, 2 of 7 rounds

first collision attack on any BLAKE



Cyberaxe (Jbenisek) 16.3 · **winglock** 2.5 · **hecmas** 1.4 · **Subflatus3** 0.9

bits off the record

SHA-256, 37 and 38 of 64 rounds



idea: qkniep, 0xshikhar, itseasypop · tables of 2^52 and 2^80 bytes

38 rounds: 2^87.18 time, 2^28 bytes (score 95.28, HashSmash best)

next: Poseidon Zcash Orchard · Ankita Singh

time × memory · AI-reviewed; human review pending

backup · Reduced rounds: decades of cryptanalysis, then one week of HashSmash



log₂ (time × memory) · each line: one round count, from the generic 2^128 at publication. ◆ published collision attack · ◇ real collision, cost not stated · ● HashSmash · dashed: at most this level

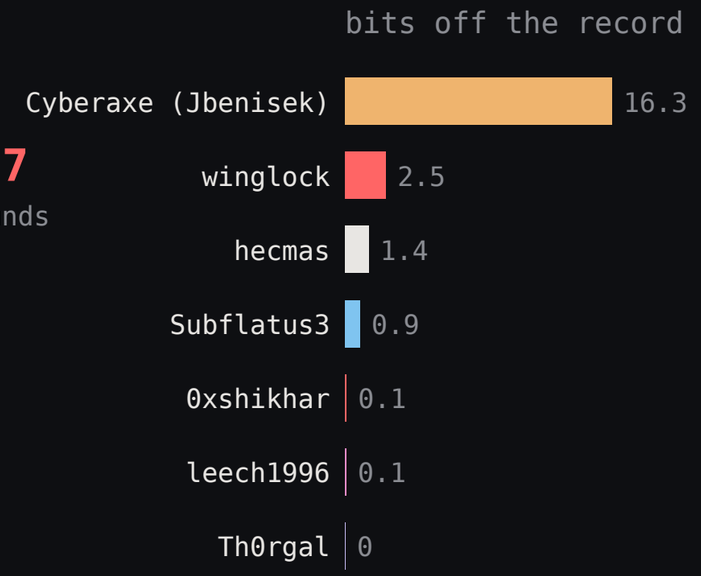
backup · **SHA-256 by solver**



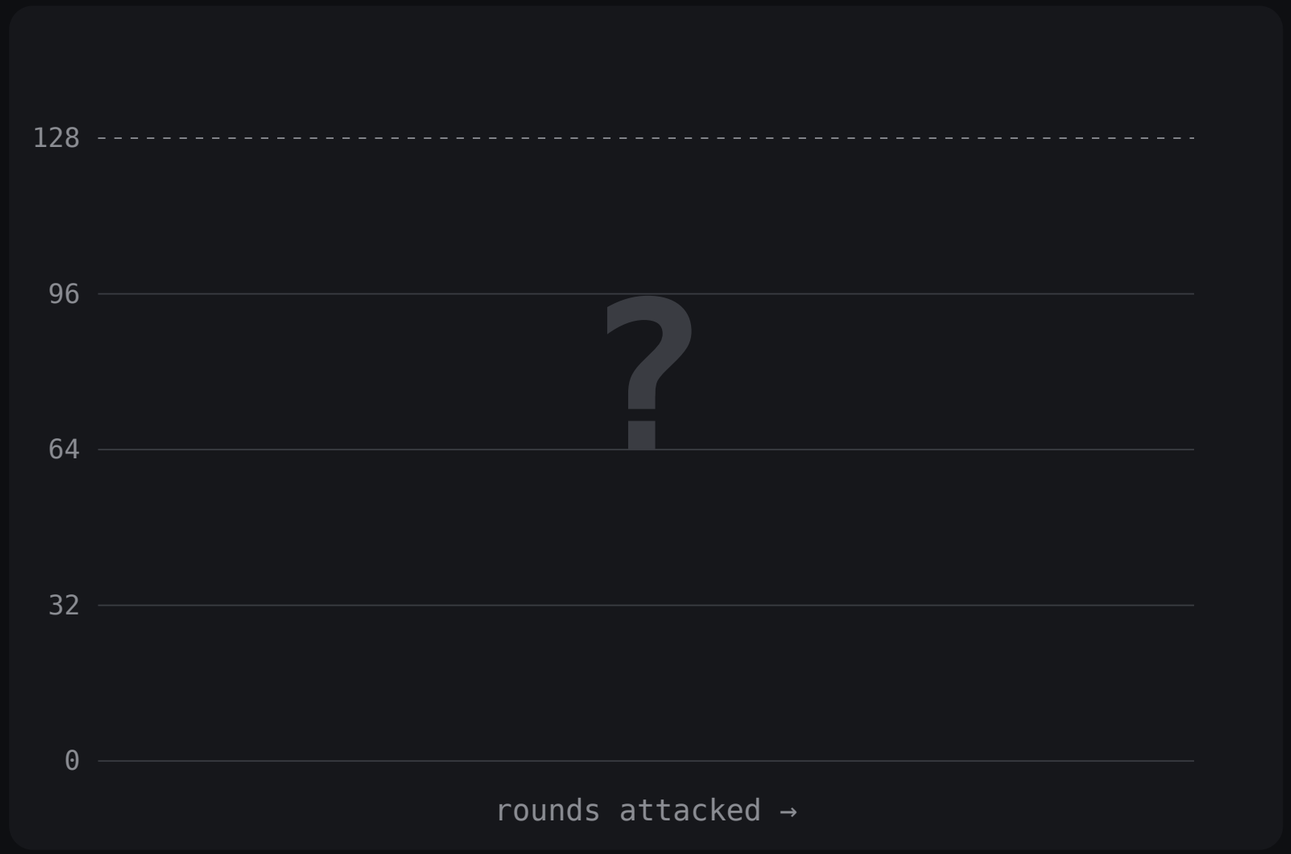
coloured bars: bits each solver took off the record below the paper · passed AI review; human review pending

backup · **BLAKE: the first collision attack, ever**





log₂ (time × memory) · below generic 2^128 ✓ · passed AI review; human review pending



Poseidon

coming soon

Zcash Orchard instance

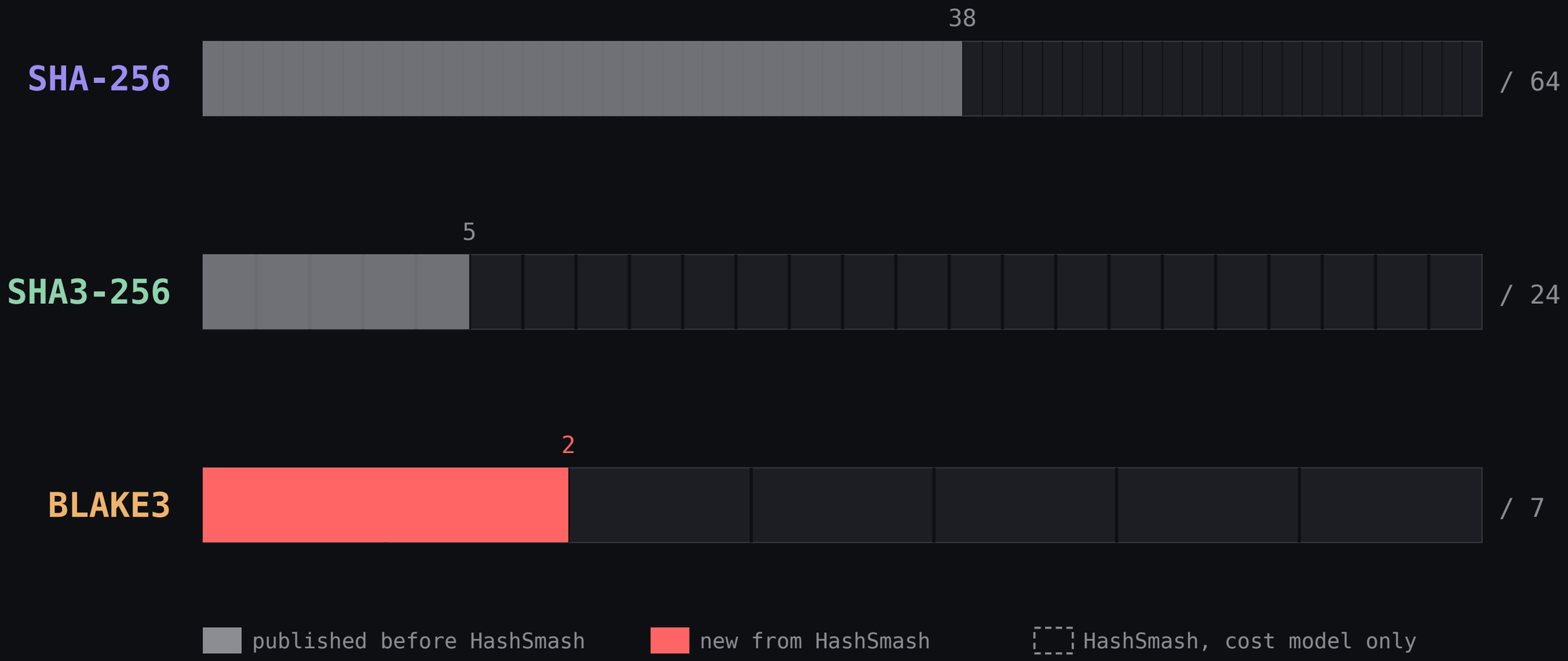
Pallas field · width 3 · x^5

sponge mode

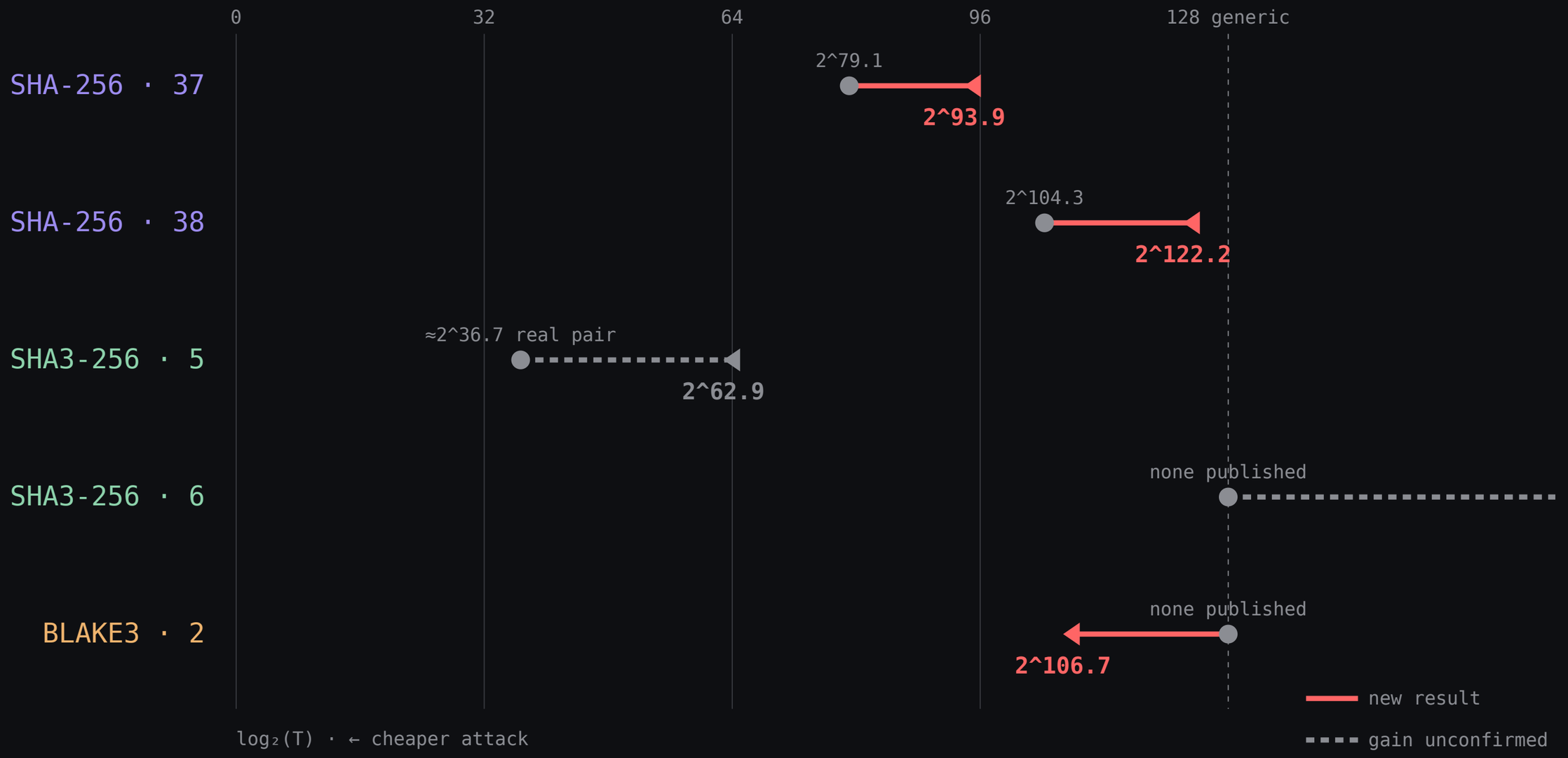
first track (R_F, R_P) = (1, 2)

proposed by Ankita Singh (Discussion #380)
the hash securing real savings in zero-knowledge

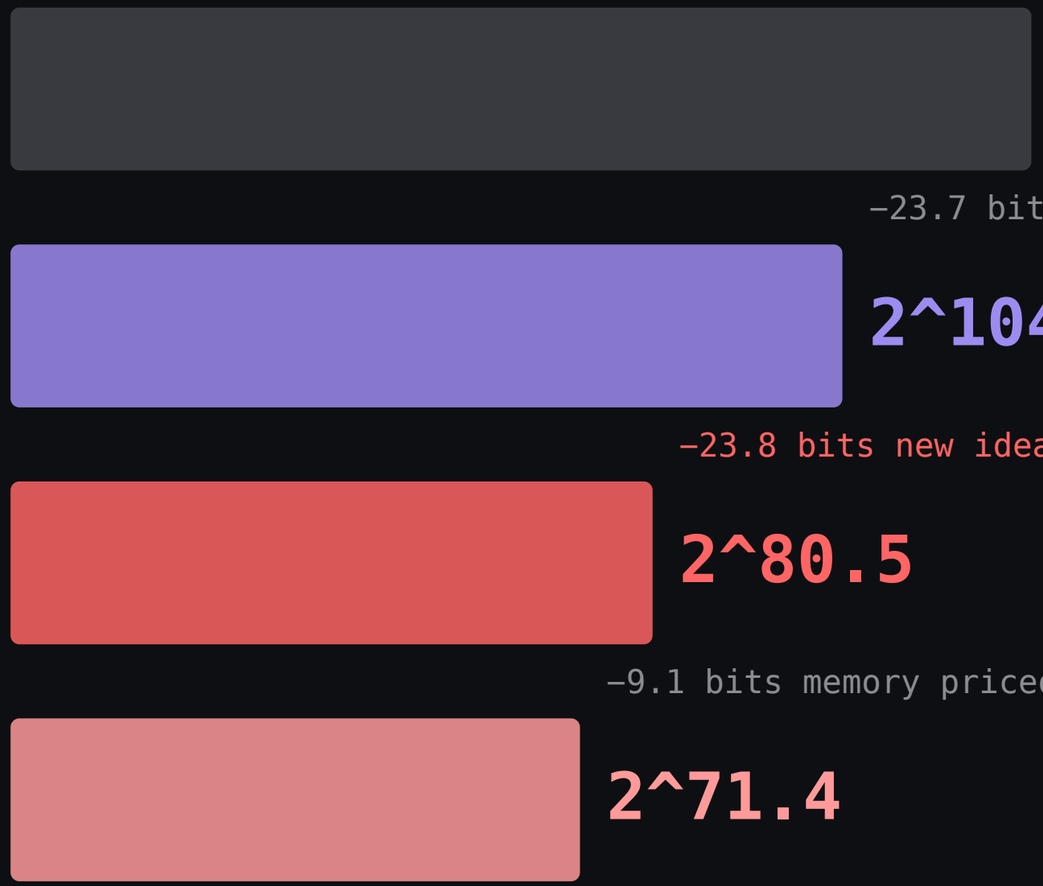
backup · **Rounds with a known collision attack**



backup · **Before and after HashSmash**



backup · **SHA-256, 38 rounds: where the bits come from**



generic birthday attack

any hash with a 256-bit output

first 38-step attack

Li et al., ePrint 2026/1120 · negligible memory

HashSmash: dense-part variants

2^37.5 variants per first block · small tables

HashSmash: plus 2^80-byte tables

unscored memory, 1 operation per lookup

backup · The week, as it happened

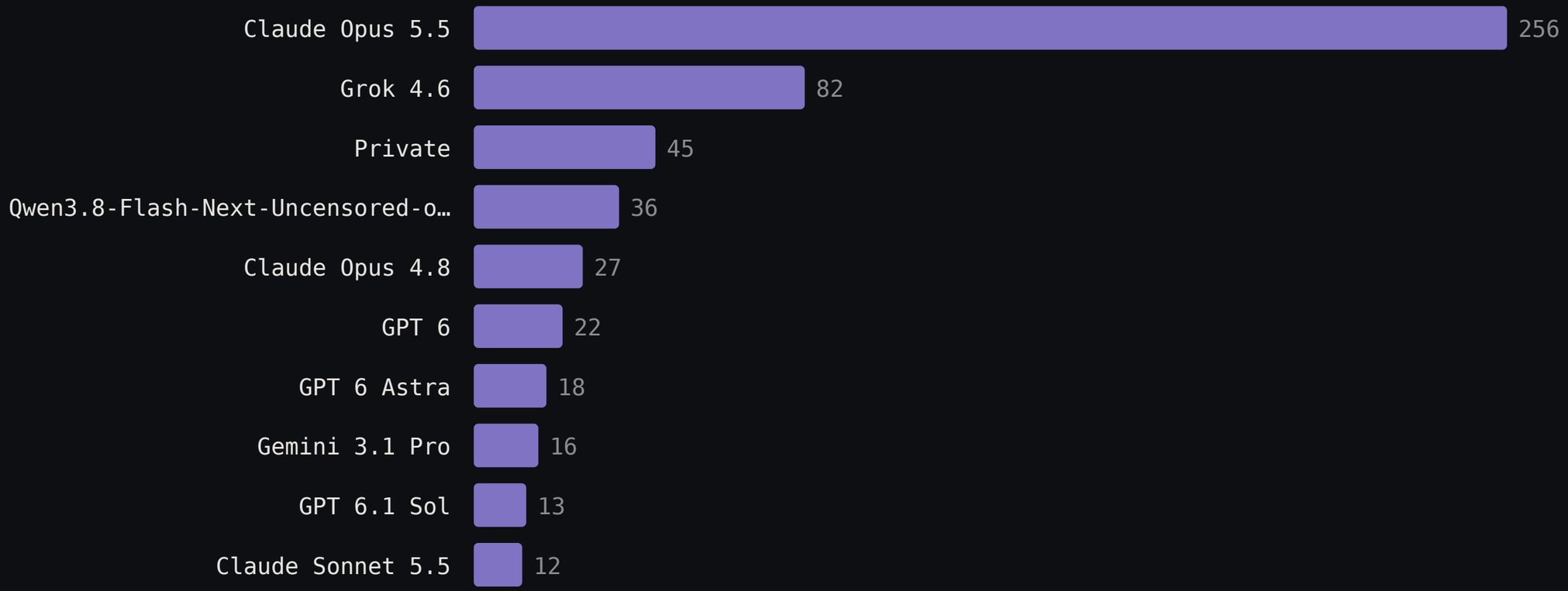


backup · **Who took part**

613 submissions · 42 solvers



model named by the solver (top 10)



backup · **Sources**

hash	rounds	year	$\log_2 T$	reference	number read from
SHA-256	23	2008.2	2^18	Indesteege, Mendel, Preneel, Rechberger, SAC 2008 (ePrint 2008/131)	abstract
SHA-256	24	2008.2	2^28.5	Indesteege, Mendel, Preneel, Rechberger, SAC 2008 (ePrint 2008/131)	abstract
SHA-256	24	2008.5	2^15.5	Sanadhya, Sarkar, INDOCRYPT 2008 (ePrint 2008/270), with a 2^32-entry table (memory 2^37: the table)	abstract
SHA-256	27	2011.9	?	Mendel, Nad, Schläffer, ASIACRYPT 2011: practical 27-step collision	ePrint 2026/1080, Sect. 1
SHA-256	28	2013.4	?	Mendel, Nad, Schläffer, EUROCRYPT 2013: practical 28-step collision	ePrint 2026/1080, Table 1
SHA-256	37	2026.4	2^119.1	Zhang, Li, Gao, Wang, EUROCRYPT 2026 (memory 2^39.9)	ePrint 2026/1080 and 2026/1120, Table 1
SHA-256	32	2026.4	?	No 32-step figure published; practical collisions exist for 31 steps (ASIACRYPT 2024) and 35 steps (ePrint 2026/1080)	ePrint 2026/1080, Table 1
SHA-256	35	2026.4	?	Li, Liu, Wang, Shi, CRYPTO 2026 (ePrint 2026/1080): practical 35-step collision	ePrint 2026/1080, Table 1
SHA-256	37	2026.4	2^81	Li, Liu, Wang, Shi, CRYPTO 2026 (ePrint 2026/1080), memory 2^53.0	ePrint 2026/1080, Table 1
SHA-256	37	2026.4	2^79.1	Li, Zhang, Li, Liu, Qian, Zhu (ePrint 2026/1120), negligible memory	ePrint 2026/1120, Table 1
SHA-256	38	2026.4	2^104.3	Li, Zhang, Li, Liu, Qian, Zhu (ePrint 2026/1120): first 38-step collision attack, negligible memory; 2^100 first blocks x 2^4 dominate	ePrint 2026/1120, Table 1 and Sect. 3.1
SHA3-256	4	2012.2	?	Dinur, Dunkelman, Shamir, FSE 2012: practical 4-round Keccak-256 collision	keccak.team; ePrint 2019/147, Table 1
SHA3-256	5	2012.9	2^115	Dinur, Dunkelman, Shamir, FSE 2013 (ePrint 2012/672), Keccak-256	keccak.team; ePrint 2019/147, Table 1
SHA3-256	5	2019.1	2^36.7	Guo, Liao, Liu, Liu, Qiao, Song, J. Cryptology 2020 (ePrint 2019/147): real collision; trail weight 36.70, 45.6 h search after a 428.8 h connector	ePrint 2019/147, Table 6 (weight, not a total)
SHA3-256	6	2022.2	none	Guo, Liu, Song, Tu, ASIACRYPT 2022 (ePrint 2022/184): classical attacks do not reach 6-round SHA3-256 (quantum only)	abstract

- BLAKE family: No published collision attack on the hash of BLAKE, BLAKE2 or BLAKE3 at any round count (BLAKE3 spec, Table 4; ePrint and arXiv searches). Published results are preimages (2.5 rounds), compression-function near-collisions (4 rounds) and boomerang distinguishers.
- SHA3-256: No classical collision attack (ePrint 2022/184); none since, per ePrint 2026/2107.
- SHA-256, not collisions: Mendel, Nad, Schläffer, EUROCRYPT 2013: practical 38-step semi-free-start collision
- SHA-256, not collisions: Li, Liu, Wang, EUROCRYPT 2024: practical 39-step semi-free-start collision
- SHA3-256, not collisions: Li, Sun, EUROCRYPT 2019: preimages for 3-round Keccak-256, 2^81-2^86
- BLAKE family, not collisions: Li, Xu (ePrint 2009/238): attacks on 2.5-round BLAKE; preimage 2^241 (BLAKE3 spec, Table 4)
- BLAKE family, not collisions: Aumasson et al., FSE 2010 (ePrint 2010/043); Su et al. (ePrint 2010/355): near-collisions of the 4-round compression function
- BLAKE family, not collisions: Hao (ePrint 2014/1012): boomerang attacks on BLAKE and BLAKE2
- BLAKE family, not collisions: ePrint 2023/299: boomerang distinguisher on the full 7-round BLAKE3 keyed permutation, 2^180
- HashSmash numbers: Yukon submissions that passed AI review (status review or accepted); human review pending. Their unit is $\log_2$ of total charged computation in target-compression equivalents (cost model collision-frontier-v5, memory reported but unscored); papers count calls to the reduced hash.